definition manually, you can take that approach. Visual Studio and the .NET tool simplify and automate some of these steps.

As in section 11.4.1, the instructions in 11.4.2 assume that you have a console app that needs to call your API, that the API is accessible, and that it has an OpenAPI description. To generate a client by using NSwag, follow these steps:

1. Ensure that the API application is running and that the OpenAPI description JSON file is accessible. Note the URL at which the JSON file is exposed. In the source code associated with the book, run the OpenApiExample project.
2. Install the .NET OpenAPI tool (http://mng.bz/GyOv) globally by running

```
dotnet tool install -g Microsoft.dotnet-openapi
```

3. From the project folder of your console app, add an OpenAPI reference by using the following command, substituting the path to the OpenAPI document and the location to download the JSON file to:

```
dotnet openapi add url http://localhost:5062/swagger/v1/swagger.json
➥ --output-file OpenAPIs\fruit.json
```

   TIP If you're running the sample code by using `dotnet run`, you can find the OpenAPI document at the preceding URL. This location is also displayed in the Swagger UI.

4. Update the packages added to your project by running the following commands from the project folder:

```
dotnet add package NSwag.ApiDescription.Client
dotnet add package Microsoft.Extensions.ApiDescription.Client
dotnet add package Newtonsoft.Json
```